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# Semi-Empirical Equation of PEMFC Considering Operation Temperature

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**Abstract:** The characteristic of a proton exchange membrane fuel cell is shown by the relationship between its output voltage and current. The output cell voltage determines the activation loss, ohmic loss, and concentration loss. Studying these losses is important to understand their effect of the reduction factor of the output of the fuel cell. One method of calculating these losses is by the curve-fitting using a semi-empirical equation. This method is useful for quickly predicting the performance. In previous studies, the fuel cell temperature has been assumed to be steady. However, the operation temperature of the fuel cell is not constant in the real fuel cell systems, such as residential and vehicular fuel-cell stacks. Therefore, it is difficult to separate the different types of losses on a polarization curve. In this study, we propose a new semi-empirical equation considering the fuel cell's operation temperature. Through thermodynamic, mechanistic, and electrochemical modeling, we express the initial voltage, activation loss, ohmic loss, and the concentration loss. We compare the fitting parameters of the proposed equation with those of equations proposed in previous studies. We also study the effects of operating temperature and cathode gas on the parameters that influence the performance.

**Keywords:** Proton exchange membrane fuel cell, modeling, overpotential, current interrupt

## 1. Introduction

The use of fuel cells is expected to help mitigate environmental problems such as fossil fuel depletion and greenhouse gas emissions. One type of fuel cell, the proton exchange membrane fuel cell (PEMFC), has the characteristics of low operation temperature and rapid startup; therefore, this type of fuel cell is used as co-generation systems and fuel-cell vehicles (FCV).

Typically, the fuel cell characteristic is shown by the relationship between its output voltage and current. It is called the  $I$ – $V$  curve or the polarization curve and is used as an index of the performance of fuel cells and stacks. Generally, the output cell voltage determines the initial voltage, the activation loss due to reaction kinetics, ohmic loss due to ionic and electronic resistance, and concentration loss due to mass transport. These losses are called overpotentials. Studying these losses is important to understand their effect on the reduction factor of the output of

the fuel cell. Many methods have been proposed for separately determining these losses using a polarization curve.

One method of calculating these losses is by curve fitting using a semi-empirical equation. This method is fitted to the measured data and the value of the equation derived theoretically and experientially. There are simple,<sup>1</sup> theoretical,<sup>2,3,4</sup> and complex equations to understand the voltage reduction factor such as cathode flooding,<sup>5,6</sup> water transport,<sup>7</sup> and CO kinetics.<sup>8</sup> Springer et al. developed a limited model of Nafion membrane.<sup>3</sup> Amphlett et al. separated the activation and ohmic loss using the proposed semi-empirical equation.<sup>4</sup> Kim et al. devised the simplest empirical equation to separate the activation, ohmic, and concentration loss based on the theory of fuel cell.<sup>1</sup> Squadrito et al.<sup>5</sup> modified the concentration loss of Kim et al.'s equation.<sup>1</sup> Pisani et al.<sup>6</sup> developed the factor of cathode flooding in the proposed equation and it is similar to Squadrito et al.'s equation.<sup>5</sup> Haji et al. devised the semi-empirical equation thought thermodynamics.<sup>2</sup> Using other methods, Gasteiger and Kocha separated the activation and concentration loss using IR-free voltage versus log-scaled current density.<sup>9,10</sup> Furthermore, the software and algorithm based on the semi-empirical equation used the prediction of the fuel cell performance.<sup>11,12,13</sup>

Although the methods by curve fitting using a semi-empirical equation cannot predict the fuel cell performance outside the operation condition, they are useful to quickly predict it.<sup>14</sup> In the methods of previous studies, the fuel cell temperature was assumed to be steady during the operation. However, it is not constant in real fuel cell systems such as residential and vehicular fuel cell stacks. In previous studies, the operation temperature

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was forced constant by the high room temperature and cooling of air and water.<sup>15</sup> Therefore, it is difficult to separate the different types of losses on the polarization curve with varying fuel cell temperature in actual operation conditions.

In this study, we propose a new semi-empirical equation considering the operation temperature of the fuel cell. Through thermodynamic, mechanistic, and electrochemical modeling, we expressed the initial voltage, the activation loss, ohmic loss, and concentration loss as a function of the operation temperature. Furthermore, we compare the fitting parameters of the proposed equation with Kim et al.<sup>1</sup> and Squadrito et al.<sup>5</sup> Moreover, the effects of the operation temperature and the cathode gas on fuel cell parameters are studied.

## 2. Methodology

### 2.1 Semi-Empirical Equation

Generally, the output cell voltage,  $V$ , is expressed as follows:

$$V = E_0 - \eta_{act} - \eta_{ohmic} - \eta_{con} \quad (1)$$

where  $E_0$  is the initial voltage.  $\eta_{act}$ ,  $\eta_{ohmic}$ , and  $\eta_{con}$  denote the activation loss, ohmic loss, and concentration loss due to mass transport, respectively.

One method of calculating these losses is by curve fitting using a semi-empirical equation. The following Eqs. (2) and (3) that we use in this study were proposed by Kim et al.<sup>1</sup> and Squadrito et al.,<sup>5</sup> respectively,

$$V = E_{OCV} - [A \ln(i)] - R_{ohmic} \cdot i - m \exp(n \cdot i) \quad (2)$$

$$V = E_{OCV} - [A \ln(i)] - R_{ohmic} \cdot i + \alpha i^k \ln(1 - \beta i) \quad (3)$$

where  $E_{ocv}$  denotes the open cell voltage, and  $i$  is the current density.  $A$  and  $R_{ohmic}$  are the fitting parameter of the activation loss and the ohmic loss, respectively.  $m$ ,  $n$ ,  $\alpha$ ,  $\beta$ , and  $k$  are the fitting parameters of the concentration loss. Especially,  $k$  is the index of the flooding proposed by Pisani et al.<sup>6</sup>;  $k = 0$  means the absence of water flooding theoretically. However, it was constant on the curve-fitting in the studies.<sup>5,6</sup> These equations have minimum factors to separate these losses when the fuel cell temperature was assumed to be steady during the operation. However, these do not have the factors of the fuel cell operation temperature for simplicity of the equation. Also, the previous study's equations<sup>4,11,13</sup> have the factors, although these are too complex and need special software. Therefore, in this study, the proposed semi-empirical equation must be simple and able to consider the operation temperature of the fuel cell.

### 2.2 Initial Voltage in Proposed Equation

For a PEMFC, the initial voltage,  $E_0$ , is equal to the thermodynamic potential. It is calculated from the modified Nernst equation<sup>2,16</sup>:

$$E_0 = E + \frac{\Delta \hat{s}}{nF} (T - T_0) - \frac{RT}{nF} \ln \left( \frac{a_{H_2O}}{a_a \cdot a_c^{1/2}} \right) \quad (4)$$

where  $E$  denotes the standard-reversible voltage;  $\Delta \hat{s}$  denotes the entropy of reaction;  $n$  is the number of electrons transferred in the reaction;  $F$  is Faraday's constant;  $T$  is the fuel cell temperature;  $T_0$  is the reference temperature;  $R$  is the universal gas constant; and  $a_{H_2O}$ ,  $a_a$ , and  $a_c$  denote the activities of water, anode gas, and cathode gas, respectively. Through thermodynamics, these parameters are obtained from the work of Haji<sup>2</sup> under the following experimental:  $E = 1.229$  V,  $\Delta \hat{s} = -163.23$  J/mol K,  $n = 2$ ,  $F = 96485.35$  C/mol,  $T_0 = 298.15$  K,  $R = 8.314$  J/mol K,  $a_{H_2O} = 1$ ,  $a_a = 0.1$ ,  $a_c = 0.21$  (Air) or 1 (O<sub>2</sub>).

### 2.3 Activation Loss in Proposed Equation

According to the Butler–Volmer equation, the activation loss,  $\eta_{act}$ , is given by

$$\eta_{act} = -\frac{RT}{\alpha nF} \ln(i_0) + \frac{RT}{\alpha nF} \ln(i) \quad (5)$$

where  $i_0$  is the exchange current density. When this equation is generalized, it reduces to the Tafel equation given below:

$$\eta_{act} = b + a \ln(i) \quad (6)$$

where  $b$  is called the Tafel slope. This equation is often used to describe the electrochemical kinetics in the PEMFC. However, if the temperature is not constant, the exchange current density,  $i_0$ , is in the form<sup>16</sup>

$$i_0 = nFf c_R^* e^{-\Delta G/RT} \quad (7)$$

where  $f$  denotes the decay rate of products.  $c_R^*$  is the reactant surface concentration, and  $\Delta G$  is Gibbs free energy. Therefore, generalizing Eq. (6),  $\eta_{act}$  can be expressed as a function of temperature as follows:

$$\eta_{act} = -aT \left\{ \ln(c) - \frac{\Delta G}{RT} - \ln(i) \right\} \quad (8)$$

$$= T[b + a \ln(i)] + \frac{\Delta G}{R} b$$

$$b \left[ T + \frac{\Delta G}{R} \right] + a \ln(i) = B + A \ln(i) \quad (9)$$

where  $a$  and  $b$  denote the fitting parameter. Gibbs free energy,  $\Delta G$ , is assumed to be 237.2 kJ/mol.<sup>17</sup>

### 2.4 Ohmic Loss in Proposed Equation

The ohmic loss,  $\eta_{ohmic}$  is given by Ohm's law in the form

$$\eta_{ohm} = R_{ohmic} \cdot i \quad (10)$$

where  $R_{ohmic}$  is the PEMFC resistance, which arises because of the electronic contribution,  $R_{elec}$ , and the ionic contribution  $R_{ionic}$ :

$$R_{ohmic} = R_{ionic} + R_{elec} = \frac{L_i}{\sigma_i S_i} + \frac{L_e}{\sigma_e S_e} \quad (11)$$

where  $\sigma_i$  and  $\sigma_e$  are the ionic and electronic conductivity. Generally, each conductivity is in the range of  $10^{-6}$ – $10^0$  S/cm and  $10^6$ – $10^8$  S/cm.<sup>16</sup> Therefore,  $R_{ionic}$  tends to dominate because ionic charge transport tends to be more difficult than electronic charge transport. In this study, we rewrite Eq. (10) as

$$\eta_{ohm} = R_{ionic} \cdot i = \frac{L_i}{\sigma_i S_i} i \quad (12)$$

Theoretically, the equation showing the relationship between the temperature and activation energy  $E_a$  is as follows<sup>16,18,19</sup>:

$$\begin{aligned} \sigma_i T &= C \exp \left[ -\frac{E_a}{RT} \right] \\ \ln(\sigma_i T) &= \ln(C) - \frac{E_a}{RT} \end{aligned} \quad (13)$$

where  $C$  is the pre-exponential factor. From previous studies at the range of cell temperature, these parameters are assumed to have the following values:  $\ln(C) = 10.8$  S/K cm,<sup>18</sup> and  $E_a = 34.6$  KJ/mol.<sup>19</sup> Therefore, the ionic conductivity is given by

$$\sigma_i = \frac{\exp(10.8)}{T} \exp^{-4162/T} \quad (14)$$

If the temperature is not constant, then  $\eta_{ohmic}$  at a given temperature,  $T$  is given as follows:

$$\eta_{ohm} = i R_{ohmic}(T) = i \frac{\sigma_{i0}}{\sigma_i} R_{ohmic}(T_0) \quad (15)$$

where  $T_0$  and  $\sigma_{i0}$  are the reference temperature and conductivity, respectively.

### 2.5 Concentration Loss in Proposed Equation

The concentration loss in the polarization losses is better described as suggested by Kim et al.<sup>1</sup>:

$$\eta_{con} = m \exp(n \cdot i) \quad (16)$$

The fitting parameter,  $m$ , is dependent on the condition inside the PEMFC. Especially, it is known to be correlated with the temperature in the same range by the previous study<sup>15</sup> as follows:

$$\begin{aligned} m(T) &= 1.1 \times 10^{-4} - 1.2 \times 10^{-6}(T - 273.15) \\ &\text{for } T > 312.15 \text{ K} \\ m(T) &= 3.3 \times 10^{-3} - 8.2 \times 10^{-5}(T - 273.15) \\ &\text{for } T < 312.15 \text{ K} \end{aligned} \quad (17)$$

### 2.6 Proposed Empirical Equation

To summarize the above, the proposed semi-empirical equation is described as

$$\begin{aligned} V &= E_{thermo}(T) - T[b + a \ln\{i\}] - b \frac{\Delta G}{R} \\ &\quad - R_{ohmic}(T) \cdot i - m(T) \exp\{n \cdot i\} \end{aligned} \quad (18)$$

We compare this equation with the equations of the previous studies discussed earlier. Subsequently, we study the effects of the operating temperature and the cathode gas on fuel cell parameters.

## 3. Experimental and Operating Conditions

A 300W-class PEMFC stack consisting of 20 cells was used in all experiments. It was cooled by the installed fan. The fuel of the anode and cathode flow field were pure hydrogen (>99.999%) and air or pure oxygen (>99.99%) with no humidification. In our experiment, hydrogen was supplied at 10 kpa by a regulator operated at the rated parameters. A purging valve, which automatically opens at intervals to purge, was used to prevent the accumulation of fuel gases and water vapor within the anode. Air was supplied using a pump, and its flow rate was measured by a flow meter. Oxygen was supplied by a regulator along with the air. A 330-kW DC electric load (Kikusui: PLZ334W) was used to load the stack in all experiments. The stack was operated at a constant current load. The cell voltages were measured using a data logger (HIOKI: 8422-50).

In all the measurements, current density was gradually increased at 2-minute intervals. Measurements were performed at the ambient temperature  $25^\circ\text{C} \pm 2$ . The stack temperature increased with the increasing current density as shown in Fig. 1. In this study, this temperature was measured by a thermocouple located at the center of the stack. Data pertaining to current density, voltage, and temperature were recorded every 2 minutes.

For fitting the measured data and the equations, we used the multiple linear regression function of Microsoft Office Excel 2010. The regression analysis tool in Excel performs linear regression analysis by using the least squares method to fit a line through a set of data.

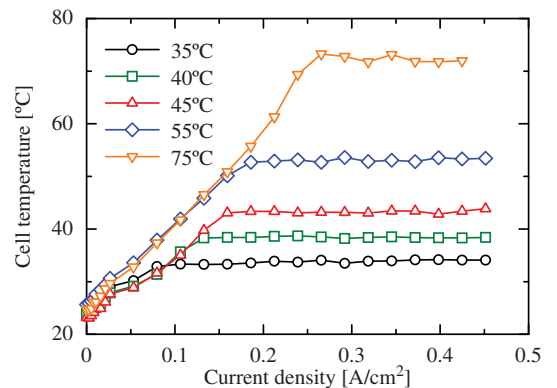


Fig. 1. Stack temperature during each operation.

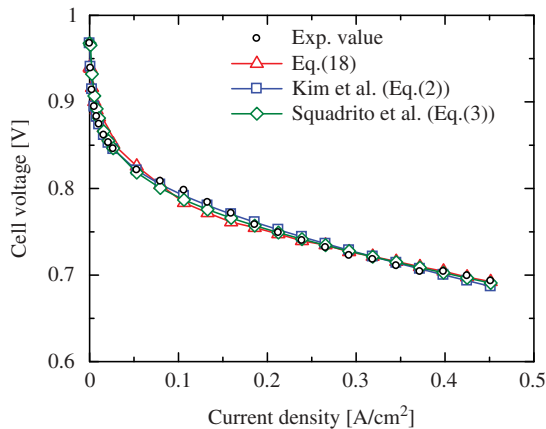
## 4. Results and Discussion

### 4.1 Comparison of Semi-Empirical Equations

Fig. 2 shows the polarization curves of the fuel cell at the center of the stack operated at 45°C. It is observed that each curve followed the path of fuel cell performance predicted by the equations. These equation models were fitted to the data and the obtained regression parameters are listed in Table 1. In addition, the table shows the internal resistance of the cell determined by the CI test.

All equations fitted the data and the correlation coefficients  $R$  were greater than 0.99. The internal resistance  $R_{ohmic}$  measured by a CI test was  $0.1441 \Omega \text{ cm}^2$  and was close to the value obtained by using the proposed equation. Typical values for  $R_{ohmic}$  are between 0.1 and  $0.2 \Omega \text{ cm}^2$ .<sup>20</sup> The fitting parameter of the activation loss  $A$  was 0.0238 V for the proposed equation and less than that obtained from the other equations. However, in a previous study,<sup>6</sup> it was 0.020–0.028 V and in the same range.<sup>5,14,15</sup> The concentration loss parameters  $m$  and  $n$  from the proposed equation and Eq. (2) by Kim et al.<sup>1</sup> were different. For Eq. (2), the concentration loss  $\eta_{con}$  was constant because  $n$  was 0 V. Therefore, this equation could not separate  $\eta_{con}$  and the other losses. On the other hand,  $\eta_{con}$  from the proposed equation was not constant because  $m$  varies with the stack temperature, and  $n$  is the only fitting parameter for this equation.

Table 2 shows the comparison with the parameters.  $D_{max}$  and  $D_{min}$  denote the max and min deviation, respectively.  $CI_{diff}$  is the difference of internal resistance between the CI test and each



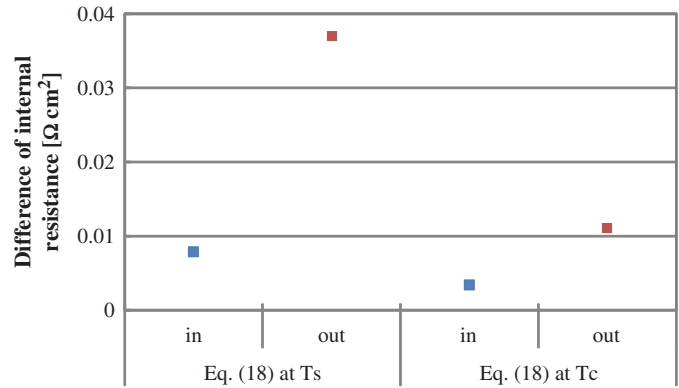
**Fig. 2.** Polarization curves obtained from each equation and from the experiment.

**Table 1.** The fitted and calculated parameters at 45°C.

|                                            | $R$   | $R_{ohmic}$<br>[ $\Omega \text{ cm}^2$ ] | $A$ [V] | $m$ [V]            | $n$ [ $\text{cm}^2/\text{A}$ ] |
|--------------------------------------------|-------|------------------------------------------|---------|--------------------|--------------------------------|
| Eq. (18)                                   | 0.995 | 0.1467                                   | 0.0238  | $5.803\text{E-}05$ | $8.329\text{E-}03$             |
| Kim et al. <sup>1</sup><br>(Eq. (2))       | 0.999 | 0.1853                                   | 0.0281  | $2.517\text{E-}02$ | 0                              |
| Squadrito et al. <sup>5</sup><br>(Eq. (3)) | 0.997 | 0.1204                                   | 0.0360  |                    |                                |
| Current interrupt                          |       | 0.1441                                   |         |                    |                                |

**Table 2.** The comparison with parameters of each equation at 45°C.

|                                         | $D_{max}$ | $D_{min}$ | $R$   | $CI_{diff}$ [ $\Omega \text{ cm}^2$ ] |
|-----------------------------------------|-----------|-----------|-------|---------------------------------------|
| Eq. (18)                                | 0.028     | 0.000     | 0.995 | 0.0026                                |
| Kim et al. <sup>1</sup> (Eq. (2))       | 0.006     | 0.000     | 0.999 | 0.0413                                |
| Squadrito et al. <sup>5</sup> (Eq. (3)) | 0.026     | 0.000     | 0.997 | 0.0237                                |



**Fig. 3.** Difference of internal resistance between the CI test and each equation.

equation value.  $D_{max}$  of Eq. (18) was larger; however,  $R$  was greater than 0.99 and  $CI_{diff}$  was considerably smaller than other equations. The deviations of Eq. (2) and (3) were smaller than Eq. (18); however,  $CI_{diff}$  was  $0.0413 \Omega \text{ cm}^2$  (28.6%) for Eq. (2), and  $0.0237 \Omega \text{ cm}^2$  (16.4%) for Eq. (3). This trend of differences was similar to the other cell as well.

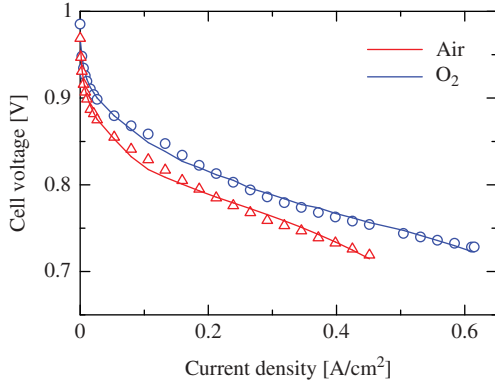
Fig. 3 shows the difference of internal resistance between the CI test and Eq. (18) at the inlet and outlet cells by each temperature. The cell temperature  $T_c$  is not equal to the operation temperature  $T_s$  because the stack is controlled by the cooling fan and equals the temperature of the center cell. The values of  $R_{ohmic}$  obtained using the cell temperature were  $0.1142 \Omega \text{ cm}^2$  and  $0.1523 \Omega \text{ cm}^2$  for the inlet and outlet cells, respectively. They were closer to the values obtained using the cell temperature determined by the CI test. Therefore, the analysis of Eq. (18) must be performed using the cell temperature or the corrected temperature if the temperature of each cell is different.

As these results, Eq. (18) was complex compared with Eq. (2) and (3). However, this equation separated these losses with more precision using the operation temperature of fuel cell.

### 4.2 Effect of the Cathode Fuel

We compared the values with air and  $\text{O}_2$  as the cathode gas at same flow rate. Fig. 4 and Table 3 summarize the effect of the cathode gas on the  $I$ - $V$  curve and the fuel cell parameters. As expected, the performance with  $\text{O}_2$  was better and Eq. (18) fitted the data well as the correlation coefficients were greater than 0.99. The internal resistance  $R_{ohmic}$  measured was  $0.1338 \Omega \text{ cm}^2$  with  $\text{O}_2$  and was observed to decrease with air as the cathode gas. This is expected as the ionic conductivity increased considerably the generated water. The fitting parameters of the activation loss





**Fig. 4.** Polarization curves for each cathode gas using the proposed equation (dots are experimental data on each cathode gas).

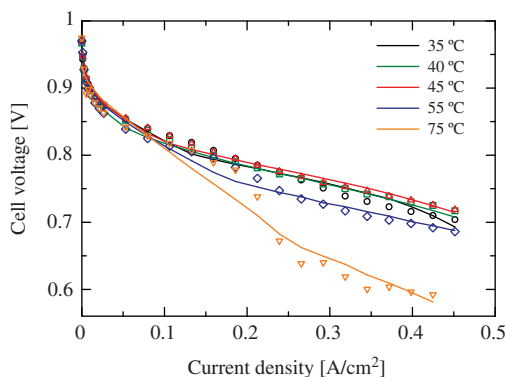
**Table 3.** The fitted and calculated parameters with each cathode gas.

| Cathod         | R     | $R_{ohmic}$<br>[ $\Omega cm^2$ ] | $a$ [V/K] | $b$ [V/K] | $n$ [ $cm^2/A$ ] |
|----------------|-------|----------------------------------|-----------|-----------|------------------|
| Air            | 0.995 | 0.1467                           | 7.484E-05 | 9.643E-04 | 8.330E-03        |
| O <sub>2</sub> | 0.997 | 0.1338                           | 5.925E-05 | 8.671E-04 | 7.463E-03        |

$a$  and  $b$  were less by 20.8% and 10.1% compared to that obtained with air owing to the increase in reactant concentration. This is also expected as the operation with O<sub>2</sub> leads to an increase in the oxygen kinetics. The fitting parameter of the concentration loss  $n$  decreased by  $0.87 \times 10^{-3} cm^2/A$  compared to that obtained with air because of the decreased reactant concentration at the catalyst surface.

#### 4.3 Effect of the Operation Temperatures

The effect of operation temperatures on the polarization curves and the fitting parameters are studied, and the results are summarized in Fig. 5 and Table 4. Generally, the performance of fuel cells is better at higher temperatures because of faster reaction kinetics that leads to lower activation loss. Thus,  $a$  was observed to decrease at 55, 75 °C as shown in Table 3. These values were



**Fig. 5.** Polarization curves at each temperature using the proposed equation (dots are experimental data at each temperature).

**Table 4.** The fitted and calculated parameters at each operation temperature.

| $T$ [°C] | R     | $R_{ohmic}$<br>[ $\Omega cm^2$ ] | $a$ [V/K] | $b$ [V/K] | $n$ [ $cm^2/A$ ] |
|----------|-------|----------------------------------|-----------|-----------|------------------|
| 35       | 0.998 | 0.1420                           | 7.711E-05 | 9.071E-04 | 1.098E-02        |
| 40       | 0.997 | 0.1449                           | 6.626E-05 | 9.174E-04 | 8.339E-03        |
| 45       | 0.995 | 0.1467                           | 7.484E-05 | 9.643E-04 | 8.330E-03        |
| 55       | 0.993 | 0.2037                           | 6.041E-05 | 9.277E-04 | 8.142E-03        |
| 75       | 0.985 | 0.5158                           | 2.890E-05 | 9.815E-04 | 1.154E-02        |

less than 19.3, 61.4%. However, the performance was no better though as shown in Fig. 5. This is the expected effect of supplied fuels such as H<sub>2</sub> and air with no humidification. Therefore,  $R_{ohmic}$  and  $n$  increased at these temperatures owing to the reduction in the moisture content of the membrane of the fuel cell. Especially,  $R_{ohmic}$  at 75 °C increased by 2.5 times compared to 45 °C. Therefore, it is important to prevent desiccation of the membrane in these temperatures using the moisture chamber. At the other temperatures, the internal resistance  $R_{ohmic}$  observed to be nearly  $0.14 \Omega cm^2$  was close to the measured value obtained by the CI test. Therefore, the moisture content of the membrane was supposed in the same range. Especially at 35 °C, the fitting parameter of the concentration loss  $n$  increased with decreasing operation temperature owing to the flooding on the catalyst surface.

#### 5. Conclusions

All equations fitted the data and the correlation coefficients were greater than 0.99. The internal resistance measured by the CI test was  $0.1441 \Omega cm^2$  and was close to the value obtained by the proposed equation. Furthermore, difference in the values of concentration loss obtained using the proposed and previously proposed equations was large because equations from previous studies could not separate  $\eta_{con}$  and the other losses.

The internal resistance measured as  $0.1338 \Omega cm^2$  at O<sub>2</sub> was observed to decrease with air because the ionic conductivity increased considerably the generated water. The fitting parameters of the activation loss decreased by 20.8% and 10.1% compared to that with air as the cathode gas owing to the increase in reactant concentration. This is also expected as operation with O<sub>2</sub> leads to an increase in the oxygen kinetics. The fitting parameter of the concentration loss  $n$  decreased by  $0.87 \times 10^{-3} cm^2/A$  compared to that with air because this could be the reason for the decreased reactant concentration at the catalyst surface.

The effect of operation temperatures on the polarization curves and the fitting parameters are studied. At 55, 75 °C,  $R_{ohmic}$  and  $n$  increased owing to the reduction in the moisture content of the membrane because of the supplied fuels such as H<sub>2</sub> and air with no humidification. At the other temperatures,  $R_{ohmic}$  was approximately  $0.14 \Omega cm^2$  and close to the value measured by the CI test. Therefore, the moisture content of the membrane of the fuel cell was probably in the same range. Especially at 35 °C, the fitting parameter of the concentration loss  $n$  increased as the

operation temperature decreased, owing to the flooding on the surface owing to its kinetics.

From the results of this study, the proposed semi-empirical equation is found to provide the conditions of the PEMFC for each type of cathode gas and at different operation temperatures. The study shows that it is useful to separate activation, ohmic, and concentration losses on the polarization curves at varying fuel cell temperatures.

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